

SALES FORECASTING USING MACHINE LEARNING

INTERNSHIP PROJECT REPORT

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Internship Organization: CODTECH IT SOLUTIONS

Duration: 6 Weeks

Submission Date: June 14, 2026

ABSTRACT

Sales forecasting is a process used to estimate future sales based on historical sales data. Accurate forecasting helps businesses make better decisions related to inventory management, budgeting, and resource planning. In this project, a Machine Learning model using the Linear Regression algorithm was developed to predict future sales values. The project demonstrates how data-driven forecasting can support business growth and planning.

INTRODUCTION

Machine Learning is widely used for predictive analysis in various business domains. Sales forecasting is one of the most important applications of Machine Learning because it helps organizations estimate future demand and improve operational efficiency.

This project uses historical monthly sales data and applies the Linear Regression algorithm to forecast future sales. The model is trained using existing sales records and predicts future sales values for upcoming months.

OBJECTIVES

- To analyze historical sales data.
- To build a Linear Regression model for sales prediction.
- To evaluate prediction accuracy using Mean Absolute Error (MAE).
- To forecast future sales values.
- To visualize sales trends using graphs.

TOOLS AND TECHNOLOGIES USED

- Python
- Google Colab
- Pandas
- NumPy
- Matplotlib
- Scikit-Learn

DATASET

The dataset consists of monthly sales values:

Month| Sales

1| 120

2| 135

3| 150

4| 170

5| 180

6| 200

7| 220

8| 230

9| 250

10| 270

11| 290

12| 310

METHODOLOGY

Data Preparation

The sales data was created and stored in a Pandas DataFrame.

Feature Selection

The Month column was selected as the input feature and the Sales column as the target variable.

Train-Test Split

The dataset was divided into training and testing datasets using the `train_test_split()` function.

Model Training

A Linear Regression model was trained using the training data.

Model Evaluation

The Mean Absolute Error (MAE) metric was used to measure prediction accuracy.

Future Forecasting

The trained model predicted sales values for Months 13, 14, and 15.

Visualization

A graph was generated using Matplotlib showing actual sales data and the regression line.

MACHINE LEARNING ALGORITHM

Linear Regression

Linear Regression is a supervised Machine Learning algorithm used for predicting continuous numerical values. It establishes a relationship between independent variables and dependent variables and generates predictions based on historical patterns.

Advantages

- Easy to implement
- Fast training process
- Suitable for forecasting
- Easy interpretation of results

IMPLEMENTATION

The implementation process included:

1. Importing required libraries.
2. Creating the sales dataset.
3. Splitting data into training and testing sets.
4. Training the Linear Regression model.
5. Predicting sales values.
6. Calculating Mean Absolute Error.
7. Forecasting future sales.
8. Displaying graphical output.

RESULTS

The model successfully predicted future sales values for upcoming months. The graph displayed both actual sales data and the regression trend line. The Mean Absolute Error (MAE) value indicated that the model was able to generate accurate forecasts based on historical sales trends.

APPLICATIONS

- Retail Sales Forecasting
- Demand Prediction
- Inventory Management
- Financial Planning
- Business Analytics

ADVANTAGES

- Improves business planning.
- Supports data-driven decisions.
- Helps estimate future demand.
- Reduces uncertainty in forecasting.
- Easy to understand and implement.

CONCLUSION

The Sales Forecasting Using Machine Learning project successfully demonstrated the use of the Linear Regression algorithm for predicting future sales. Historical sales data was analyzed to train the model, and future sales values were forecasted accurately. The project highlights the practical application of Machine Learning in business forecasting and decision-making.

REFERENCES

1. Python Official Documentation

2. Pandas Documentation
3. NumPy Documentation
4. Matplotlib Documentation
5. Scikit-Learn Documentation

END OF REPORT